

Air Quality Prediction Using Environmental Sensor Data

**Ana Maria Gonzalez Villa
Juan David Hernández Florez
Liceth Katherine Forero Tusarma
Wagner Alexander Moreno Alvarado**

DAMO 510: Predictive Analytics

Agenda

- 1. Business Understanding**
- 2. Data Understanding**
- 3. Data Preparation**
- 4. Modeling**
- 5. Evaluation**
- 6. Deployment and Reflection**
- 7. Conclusions**



1. Business Understanding

Problem Statement

- CO is harmful & invisible
- Monitoring is expensive
- Need alternative estimation methods

How well sensor data, environmental variables and temporal information can predict real pollution levels.



1. Business Understanding

Project Objective



**Predict
CO(GT)**

Using:

- Sensor data
- Weather variables
- Time features

2. Data Understanding

- 9,357 observations
- 15 variables
- Italy (2004–2005)

	Reference Pollutant Measurements
	Sensor Responses
	Weather conditions (Temperature, Relative Humidity,

Date	Time	CO(GT)	PT08.S1(CO)	NMHC(GT)	C6H6(GT)	PT08.S2(NMHC)	NOx(GT)	PT08.S3(NOx)	NO2(GT)	PT08.S4(NO2)	PT08.S5(O3)	T	RH	AH
10-03-04	18:00:00	2.6	1360	150	11.9	1046	166	1056	113	1692	1268	13.6	48.9	0.7578
10-03-04	19:00:00	2	1292	112	9.4	955	103	1174	92	1559	972	13.3	47.7	0.7255
10-03-04	20:00:00	2.2	1402	88	9.0	939	131	1140	114	1555	1074	11.9	54.0	0.7502
10-03-04	21:00:00	2.2	1376	80	9.2	948	172	1092	122	1584	1203	11.0	60.0	0.7867
10-03-04	22:00:00	1.6	1272	51	6.5	836	131	1205	116	1490	1110	11.2	59.6	0.7888
10-03-04	23:00:00	1.2	1197	38	4.7	750	89	1337	96	1393	949	11.2	59.2	0.7848
11-03-04	0:00:00	1.2	1185	31	3.6	690	62	1462	77	1333	733	11.3	56.8	0.7603
11-03-04	1:00:00	1	1136	31	3.3	672	62	1453	76	1333	730	10.7	60.0	0.7702
11-03-04	2:00:00	0.9	1094	24	2.3	609	45	1579	60	1276	620	10.7	59.7	0.7648
11-03-04	3:00:00	0.6	1010	19	1.7	561	-200	1705	-200	1235	501	10.3	60.2	0.7517
11-03-04	4:00:00	-200	1011	14	1.3	527	21	1818	34	1197	445	10.1	60.5	0.7465
11-03-04	5:00:00	0.7	1066	8	1.1	512	16	1918	28	1182	422	11.0	56.2	0.7366
11-03-04	6:00:00	0.7	1052	16	1.6	553	34	1738	48	1221	472	10.5	58.1	0.7353
11-03-04	7:00:00	1.1	1144	29	3.2	667	98	1490	82	1339	730	10.2	59.6	0.7417
11-03-04	8:00:00	2	1333	64	8.0	900	174	1136	112	1517	1102	10.8	57.4	0.7408
11-03-04	9:00:00	2.2	1351	87	9.5	960	129	1079	101	1583	1028	10.5	60.6	0.7691

Target Variable

3. Data Preparation

- **Missing values** = -200 → removed
(Represents sensor malfunction or invalid readings)
- **Final dataset:** 827 rows
- **Feature engineering:**
Hour (intra-day patterns)
Weekday (weekly patterns)

Ensures data reliability
Captures real-world pollution cycles



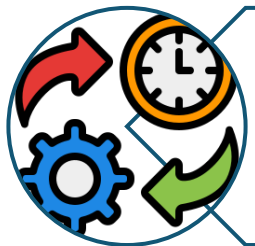
Exploratory Insights



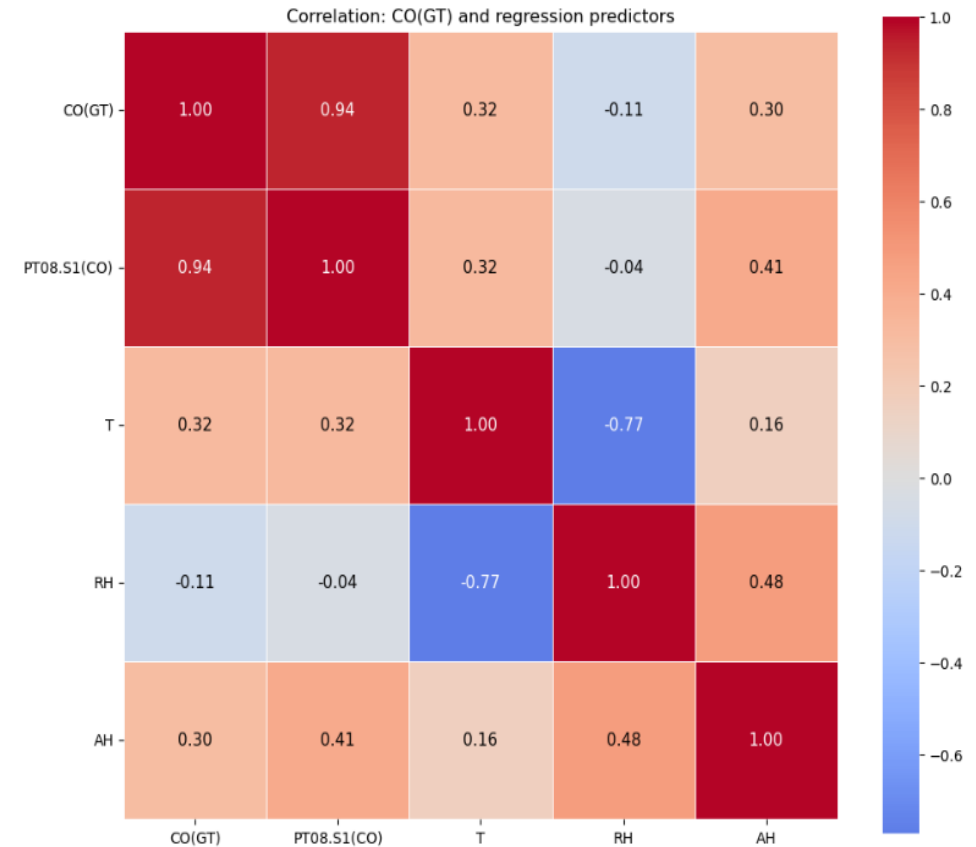
Strong correlation:
PT08.S1(CO) → CO (0.94)
→ Main predictor



Multicollinearity: T, RH, AH
Redundant information →
Affects model stability

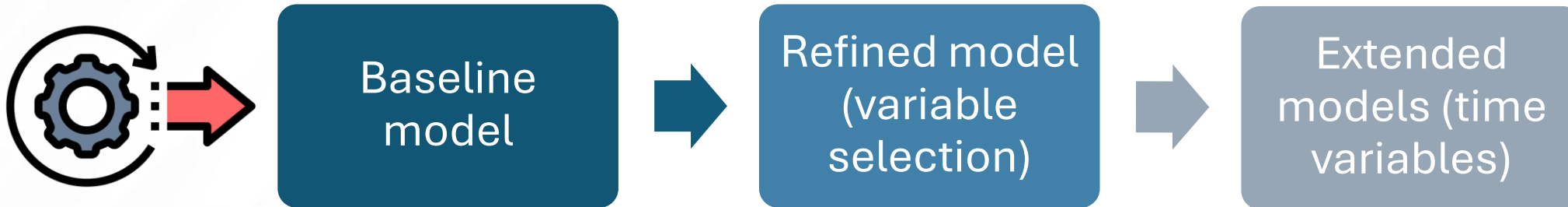


CO varies by time
Influenced by traffic & daily
activity cycles



4. Modeling

- Goal: Explain CO(GT) variation
- Step-by-step regression:



Focus on:

- Statistical significance
- Multicollinearity (VIF)
- Model stability and interpretability

Model 1: Baseline Model

$$R^2 \approx 0.886$$

Variables included:

PT08.S1(CO) → CO
sensor reading

T (Temperature) →
Ambient
temperature (°C)

**RH (Relative
Humidity)** → Air
moisture (%)

**AH (Absolute
Humidity)** → Total
water vapor in air

Key findings:

- PT08.S1(CO) → strong & significant
- T, RH, AH → not significant

Issues:

- High multicollinearity (VIF > 10)



Model 2: Refined Model

(Without AH)

$R^2 \approx 0.886$



Variables included:

PT08.S1(CO) → CO sensor reading

T (Temperature) → Ambient temperature (°C)

RH (Relative Humidity) → Air moisture (%)

* **AH removed due to: High VIF, Lack of significance**

Key results:

- T and RH → now significant
- Stable coefficients

Improvements:

- $VIF < 5$ (no multicollinearity)
- Lower condition number
- More stable coefficients

Model 3: Extended Model



$$R^2 \approx 0.888$$

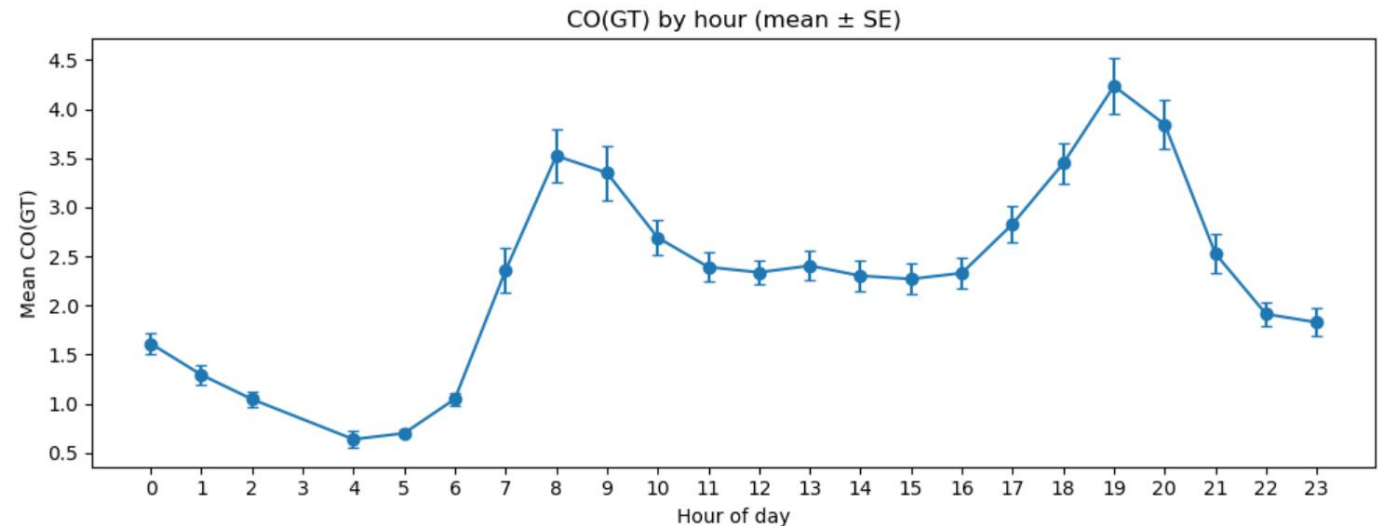
Purpose: Capture daily pollution patterns

Results:

- Hour → significant
- CO peaks (morning & evening)

Insights:

- Temporal patterns matter
- Sensor still dominant



Model 4: Extended Model



(WeekDay)

$$R^2 \approx 0.891$$

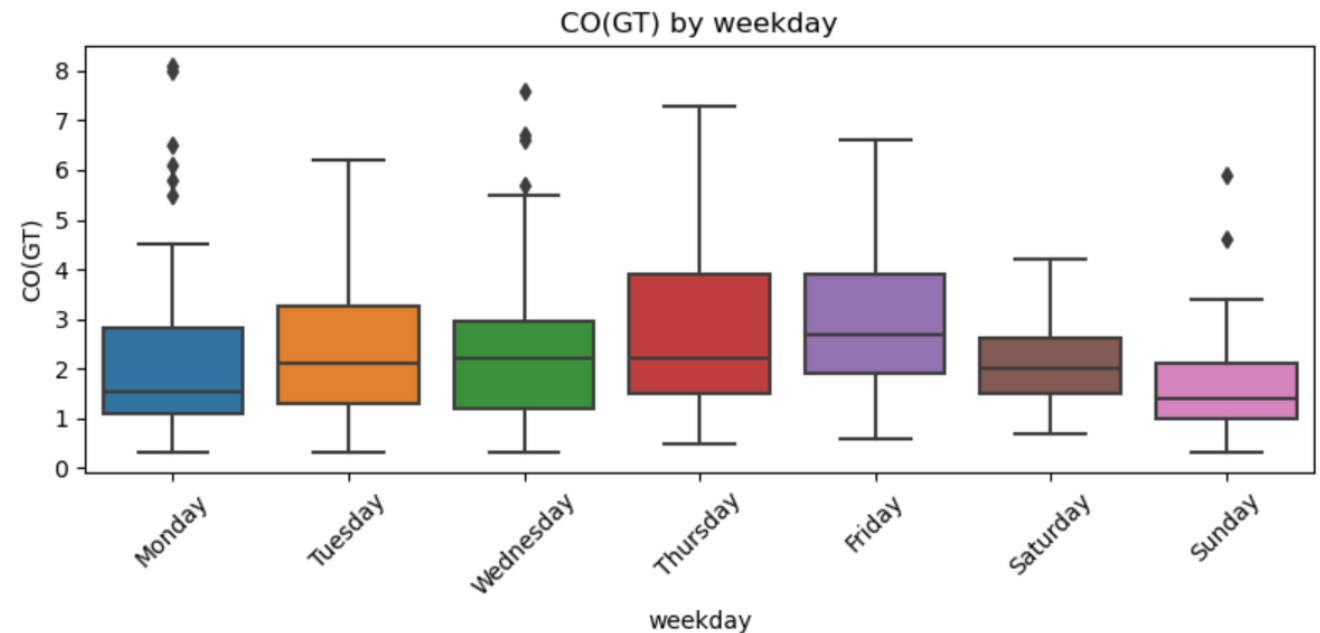
Weekday dummy variables added

Findings:

- Lower CO levels on weekends
- Some weekdays significant

Impact:

- Small improvement in
- Limited additional value



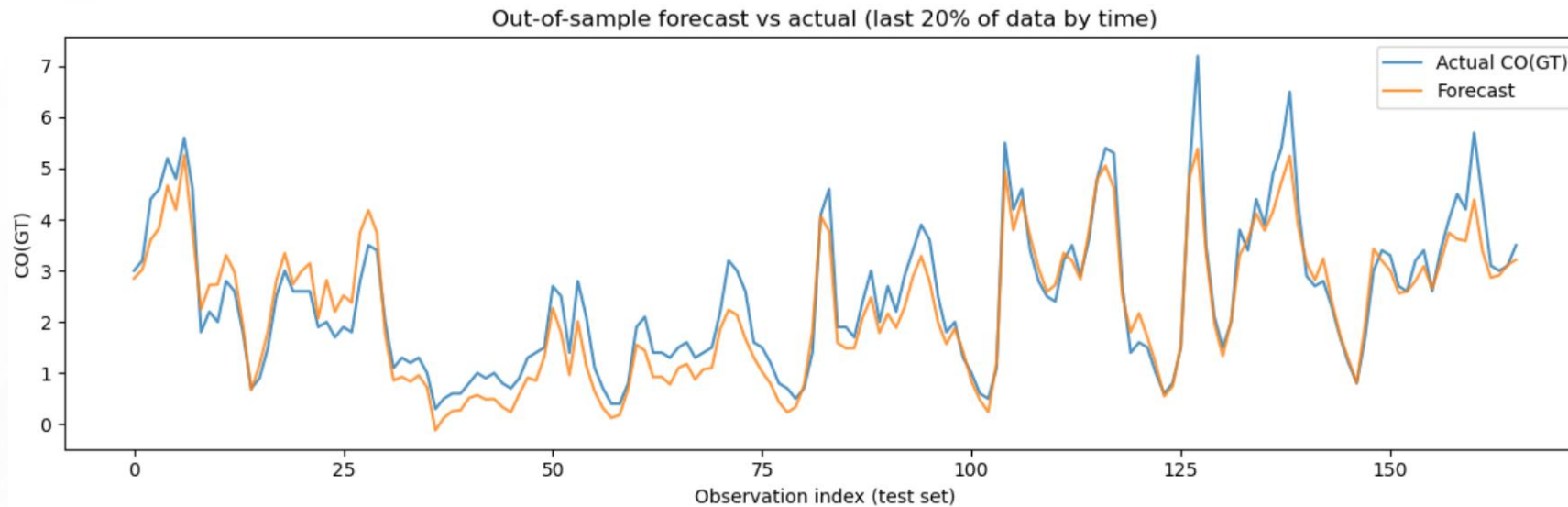
Out-of-Sample Forecasting

Performance:

- $MAE \approx 0.39$
- $RMSE \approx 0.47$
- $MAPE \approx 20.8\%$

Results:

- Good overall fit on unseen data
- Underestimates peaks



5. Evaluation

- R^2 : 0.886 $\rightarrow$ 0.891

Key evaluation insights:

- Strong explanatory power
- Consistent performance across models
- Small improvement from time variables



High explanatory power
Limited gain from added complexity



Model Comparison & Insights

- Sensor = dominant predictor
- Hour captures daily pollution cycles
- Removing AH improved stability
- Weekday = small effect

Model	R ²	MAE	RMSE	Interpretation
Model 2: Refined	0.886	0.3583	0.4758	Stable and easy to interpret
Model 3: With hour	0.888	0.3541	0.4713	Capture better daily pattern
Model 4: With weekday	0.891	0.3583	0.4764	Better fit, small improvement



Performance improves slightly, but structure remains stable

Reliability & Limitations

- $VIF < 5$ after removing AH
- Good fit (actual vs predicted)
- Underestimates peaks
- Sample reduced (827 rows)



Reliable overall, weaker for extreme pollution peaks

Final Model Selection



- Highest R^2 : model with time variables
- Best interpretability: Refined model
- Trade-off:

Accuracy vs Simplicity



Refined model = best practical choice

6. Deployment and Reflection

Practical Applications

IoT Sensor Applications in Smart Cities: Project-Based Review



- Low-cost monitoring
- Smart cities
- Early warning systems
- Peak-hour alerts

Limitations & Future Work



- **Old dataset**
- **Single location**
- Underestimates extreme pollution peaks
- **Future:**
 - Advanced ML models (RF, XGBoost)
 - Time series models (ARIMA, SARIMA)

7. Conclusion

- **CO can be predicted accurately**
- **Sensor = Key driver**
- **Model is simple & effective**

